

Figure and Ground in Counterfactuals

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"It is a crucial fact about sentence topics, that equivalent sentences may have different topics." — Reinhart (1981:58).

Goodman (1947) observed: there appear to be counterfactuals with

- logically equivalent antecedents
- the same consequent, and
- different truth values.

Goodman's own examples differ slightly, and allow that both are vacuously true. I reformulated them to avoid the issue.

- (1) a. If I were Julius Caesar, I would have lived in Ancient Rome.
- b. If Julius Caesar were me, I would have lived in Ancient Rome.
- (2) a. If New York City were in Georgia, then New York City would be in the South.
- b. If Georgia included New York City, then New York City would be in the South.

The (a)-sentences seem fine, while the (b)-sentences do not. This appears to challenge Substitution.

Substitution. For any sentences p and q , if p and q are logically equivalent (true in all the same worlds), then for any sentence r , 'if p , would r ' and 'if q , would r ' are logically equivalent too.

- "I am Julius Caesar" $\equiv$ "Julius Caesar is me"
- "NYC is in Georgia" $\equiv$ "Georgia includes NYC"

Intuitive support for Substitution:

- (3) a. If Juan visited New York City, he'd see Central Park.
- b. If Juan visited the Big Apple, he'd see Central Park.
- (4) a. If Polgar won against Anand, Polgar would celebrate.
- b. If Anand lost against Polgar, Polgar would celebrate.
- (5) a. If Zara hugged Cyril, he would blush.
- b. If Cyril was hugged by Zara, he would blush.

The (a)- and (b)-sentences in each pair seem equivalent.

Why care about Substitution?

1. A wide range of theories of conditionals validate it.
2. It is a consequence of possible worlds semantics (Lewis 1970, Partee 1977), whereby the meaning of a sentence is determined by the possible worlds in which it is true.

e.g. Stalnaker (1968), Lewis (1973a), Kratzer (1981), Pollock (1976), Bennett (1984). Exceptions include Fine (2012), Ciardelli et al. (2018).

Goals of this talk

1. Show that Goodman's examples are the tip of the iceberg.

Similar challenges to Substitution arise from a wide range of expressions featuring a **subject** and an **object**, including:

- binary relations *resembles, is similar to*
- comparatives *taller than, older than*
- equatives *as tall as, as old as*
- quantifiers *every, no*
- prepositions *on top of, beneath*
- kinship terms *sister, cousin*

2. Argue against three possible responses.

- (i) Replace possible worlds with truthmakers (Fine 2012).
- (ii) The de re reply: subjects are interpreted de dicto, objects de re.
- (iii) Variation in counterpart strictness: subjects use a looser counterpart relation than objects (Lewis 1973a).

3. Give a positive proposal: an aboutness theory of counterfactuals

The primitive notion is not similarity but aboutness.

When we interpret a counterfactual antecedent, we change the part of the world its antecedent is about.

Subject–object contrasts

- (6) *Alice is 25, Bob is 15. One must be over 21 to enter the bar.*
 - a. If Alice were younger than Bob, they could both enter the bar.
 - b. If Bob were older than Alice, they could both enter the bar.
- (7)
 - a. If Alice were as young as Bob, they could both enter the bar.
 - b. If Bob were as old as Alice, they could both enter the bar.
- (8) *Alice is at home and Bob is outside in the rain. See Figure 1.*
 - a. If Alice were near Bob, they would both be dry.
 - b. If Bob were near Alice, they would both be dry.
 - c. If Alice and Bob were near each other, they would both be dry.
- (9) *Adonis is handsome, Socrates is not.*
 - a. If Socrates resembled Adonis, they would both be handsome.
 - b. If Adonis resembled Socrates, they would both be handsome.
 - c. If Adonis and Socrates resembled each other, they would both be handsome.



Figure 1: Compare 'if Alice were near Bob' with 'if Bob were near Alice'.

- (10) a. If Bach were similar to Black Sabbath, Bach would be played less at church.
 b. If Black Sabbath were similar to Bach, Bach would be played less at church.
 c. If Bach and Black Sabbath were similar, Bach would be played less at church.
- (11) *See Figure 2.*
 a. If block A were on top of block B, the blocks would form a straight line.
 b. If block B were beneath block A, the blocks would form a straight line.
- (12) *Amy has harsh parents, Beth has nice parents.*
 a. If Amy were Beth's sister, they would both be happy.
 b. If Beth were Amy's sister, they would both be happy.
- (13) *Point A is at 2, point B is at 4, so their midpoint is at 3 (Figure 3).*
 a. If A were to the right of their midpoint, A would be to the right of 4.
 b. If B were to the left of their midpoint, A would be to the right of 4.
- (14) *Suppose there are only two colours, light and dark, and only two shapes, square and circle. See Figure 4.*
 a. If every circle were dark, ...
 b. If every light shape were square, ...
- (15) *Context 1: we are told that $m = n + k$.
 Context 2: we are told that $n = m - k$.
 If n were 4, m would be 7.*

Beyond counterfactuals

- (16) *Decorators are moving furniture according to their client's instructions. Currently, the bed is on the left and the painting is on the right (see Figure 5). The client wants both to be on the left. The manager instructs the employee:*
 a. The client wants the painting to be above the bed.
 b. The client wants the bed to be below the painting.
- (17) *The client is looking at their bedroom. They say:*
 a. I wish the bed were below the painting.

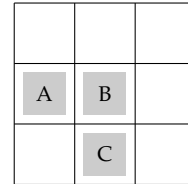


Figure 2: Compare 'if A were on top of B' with 'if B were below A'.

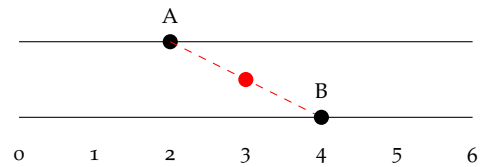


Figure 3: Point A is at 2, point B is at 4, so their midpoint is at 3.

I first learnt of subject-object contrasts in quantifiers from Benjamin Spector; see Spector and Mouly (2025), who refer to the contrast in (14) as 'restrictor constancy'.

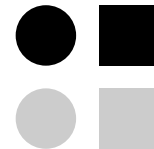


Figure 4: Compare 'if every circle were dark' with 'if every light shape were square'.

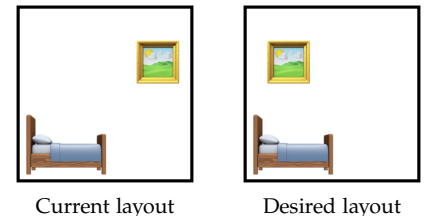


Figure 5: Redecorating scenario.

- b. I wish the painting were above the bed.
- (18) It's so cold in New York City!
 - a. I wish New York City was in Georgia.
 - b. I wish Georgia included New York City.
- (19) *The client has inspected the current layout and is not happy.*
 - a. The client is unhappy because the painting is not above the bed.
 - b. The client is unhappy because the bed is not below the painting.
- (20) *Georgia has a warmer climate than New York City, allowing Georgia to grow tasty peaches.*
 - a. New York City has bad peaches because it's not in Georgia.
 - b. New York City has bad peaches because Georgia does not include it.

Existing challenges to Substitution

1. A case from Fine (2014):

See also Fine (2012). Goodsell (2022) presents a similar example.

- (21) *Eve is in Alternative Eden, featuring one poison apple a_0 and infinitely many safe apples $a_1, a_2, \dots$.*
 - a. If Eve ate infinitely many of the apples $a_1, a_2, \dots$, she would be fine.
 - b. If Eve ate infinitely many of the apples $a_0, a_1, \dots$, she would be fine.

The antecedents are logically equivalent.

A single apple cannot separate the finite from the infinite.

2. A second case (inspired by Fine 2012).

If one disagrees that the antecedents are logically equivalent, note that they counterfactually imply each other, and are therefore predicted to be equivalent assuming Reciprocity/CSO.

- (22) *A passcode has 8 binary digits, with a space between each pair of digits. The code is currently set to 00 00 00 00. The correct code is 00 00 00 11.*
 - a. If the last two digits were 1, the code would be correct.
 - b. If for some space, every digit after it were 1, the code would be correct.

3. Experiment on Substitution (joint work with Ivano Ciardelli and Tomasz Klochowicz).

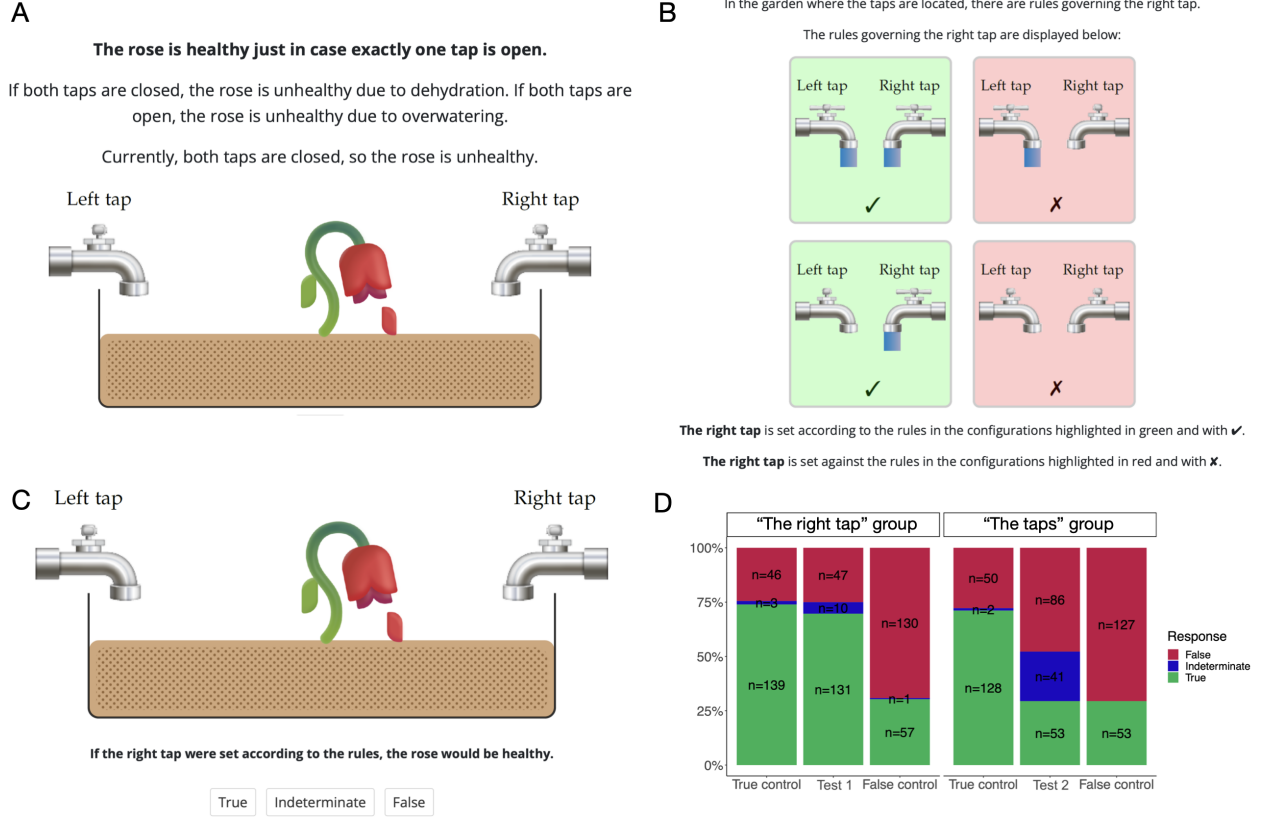


Figure 6: An experiment on Substitution. A: the scenario. B: the rules governing the scenario. C: the key test item. D: the results.

Responses

From possible worlds to truthmakers

Fine (2017) introduces a set of states, and a primitive *exact verification* between states and statements, with the requirement that a statement's exact verifiers be 'wholly relevant' to the statement's truth.

$A > C$ is true at a world w just in case for every exact verifier t of A and possible outcome u of t at w , u contains an exact verifier of C .

(Fine 2012:237)

This invalidates Substitution, but validates *Exact* Substitution.

Exact Substitution. If p and q are exactly equivalent (have the same exact verifiers), then for any r , 'if p , would r ' is true just in case 'if q , would r ' is true.

Recall from (11): 'if block a were on top of block b ' is not intersubstitutable *salva veritate* with 'if block b were beneath block a '.

Suppose that a given block a is on top of another block b . Then there is a certain state of affairs s_1 , we may describe as the state of a 's being on

top of *b*. There is also a certain state of affairs s_2 that may be described as the state of *b*'s being beneath *a*. Yet surely the states s_1 , and s_2 are the same. There is a single state of affairs *s* "out there" in reality, consisting of the blocks *a* and *b* having the relative positions that they do; and the different descriptions associated with s_1 , and s_2 would merely appear to provide two different ways at getting at this single state of affairs.

(Fine 2000:3)

the state of *a*'s being adjacent to *b* is surely the same as the state of *b*'s being adjacent to *a*.

(Fine 2000:17)

- (23) *Alice has little money. Her house is in an affordable area and her office is in an expensive area.*
- a. If Alice's house were adjacent to her office, her rent would be higher.
 - b. If Alice's office were adjacent to her house, her rent would be higher.
 - c. If Alice's office and house were adjacent, her rent would be higher.

An argument from reciprocals.

- (24) a. **Reciprocal:** Alice, Bob, and Charlie agreed.
 b. **Binary forms:** Alice agreed with Bob, Bob agreed with Charlie, Charlie agreed with Alice.
- (25) a. Alice, Bob, and Charlie are similar.
 b. Alice is similar to Bob, Bob is similar to Charlie, and Charlie is similar to Alice.

Reciprocal-first approach: treat the reciprocal as primary and derive the binary form from it.

(Lakoff and Peters 1969, Lasnik 1995, Winter 2018)

E.g. analyse '*A* agrees with *B*' as: *A* and *B* agree.

Relation-first approach: treat the binary form as primary and derive the reciprocal from it.

(Gleitman 1965, Dimitriadis 2008, Rubinstein 2009, Siloni 2012)

E.g. analyse '*A* and *B* agree' as: *A* agrees with *B*.

Problem: this only works with two arguments. With more than two, it is not at all clear how to derive the common feature from compositions of binary relations.

If the ontology of natural language serves as a guide to the structure of our conceptual life, this suggests there is no state of Alice agreeing with Bob that is not also a state of Bob agreeing with Alice.

(Fodor 1987, Pietroski 2018, Moltmann 2019). Indeed, this assumption is a key tenet of cognitive linguistics (Fodor 1975, Lakoff 1987, Croft and Cruse 2004).

The de re reply

I have heard of a touchy owner of a yacht to whom a guest, on first seeing it, remarked, "I thought your yacht was larger than it is"; and

the owner replied, “No, my yacht is not larger than it is.” What the guest meant was, “The size that I thought your yacht was is greater than the size your yacht is”; the meaning attributed to him is, “I thought the size of your yacht was greater than the size of your yacht.”

(Russell 1905:489)

For every world w' compatible with the owner's beliefs at w , the yacht's size in w' is larger than its size in w (*de re*) / w' (*de dicto*).

Here I express this using world indices (Percus 2000, Keshet 2010).

- (26) a. Alice thinks that everyone who is inside is outside.
b. If everyone who is inside were outside, the room would be empty.
- (27) *Alice believes that Bob resembles the actor Rami Malek. Alice isn't aware that Rami has an identical twin brother Sami Malek—she has no idea that Sami exists.*
Alice believes that Bob resembles Sami Malek.
- (28) a. If Socrates resembled Adonis, ...
If $\{w' : \text{Socrates' appearance in } w' \text{ is sufficiently close to Adonis' appearance in } w\}$ were true ...,
b. If Adonis resembled Socrates, ...
If $\{w' : \text{Adonis' appearance in } w' \text{ is sufficiently close to Socrates' appearance in } w\}$ were true, ...

where w is the world of evaluation.

Proposal: object terms in counterfactual antecedents are interpreted de re by default.

Challenges for the de re reply

1. Some terms seem to lack the required de re readings altogether.
Recall (12), showing that *sister* exhibits a subject–object contrast.

- (29) *John believes that Amy's parents are Rita and Denis, and that Beth's parents are Vivienne and Michael. In fact it is the reverse: Amy's parents are Vivienne and Michael, while Beth's parents are Rita and Denis.*
John believes that Amy is Beth's sister.

This seems to have no true reading. Read de re, it would say that the parents John *believes* Amy to have are the parents Beth *in fact* has.

$\forall w' \in Bel_w(j) \exists x, y (P_{w'}(a, x, y) \wedge P_w(b, x, y))$

Compare it with:

$Bel_w(x)$ are x 's belief worlds at w .
 $P_w(x, y, z)$ says that x 's parents are y and z at w .

- (30) John believes that Beth's parents are Amy's parents.
- (31) *Alice works for a bank. Bob works as a teacher. Alice hates her job and wants to be a teacher. Bob likes his job.*
a. If Alice were Bob's colleague, they would both like their jobs.
b. If Bob were Alice's colleague, they would both like their jobs.

- (32) *John believes that Alice works at a school. In fact she works at the bank. He has never met Bob before, and has no idea who he is. We know that Bob works at that school.*
John believes that Alice is Bob's colleague.

It seems impossible to get a true reading here. Compare it with:

- (33) John believes that Alice works where Bob works.

This seems to hold for all relational nouns whatsoever: sibling (share a parent), cousin (share a grandparent), colleague (share a work-place), neighbour (share a neighbourhood), collaborator (share a project), ally (share a cause), and the mate- series (e.g. band-/class-/crew-/house-/lab-/office-/play-/room-/ship-/school-/teammate).

2. Some cases have a de dicto reading that invalidates Substitution.

- (13) *Point A is at 2, point B is at 4, so their midpoint is at 3 (Figure 3).*
a. If A were to the right of their midpoint, A would be to the right of 4.
b. If B were to the left of their midpoint, A would be to the right of 4.

- Read de re, they are both not true/indeterminate, and hence have the same status.
- Read de dicto, Substitution predicts them to be equivalent.

Intuitively, however, (13a) and (13b) are not equivalent.

3. It is questionable that objects in counterfactual antecedents are interpreted de re by default.

- (35) Lucy's family are taking a family photo. Lucy is currently blocking the person behind her from being in the photo.
a. If Lucy were sitting next to Lucy, everyone would be in the photo.
b. If Lucy were sitting next to where Lucy is currently sitting, everyone would be in the photo.

If one objects to a repeated name, suppose Lucy is the birthday girl and consider:

- (34) If Lucy were sitting next to the birthday girl, everyone would be in the photo.

There is a clear contrast between (35a) and (35b). This is unexpected if 'sitting next to Lucy' is read de re by default.

- (36) If Adonis resembled Socrates and Socrates were handsome, they would both be handsome.

(36) is most naturally judged true. But if 'Adonis resembled Socrates' is judged de re by default, we would expect it to mean 'Adonis resembled Socrates' actual appearance'.

4. It violates independently plausible assumptions about the availability of de re readings.

Principle of Charity. When multiple readings of a sentence are available, some of which are true, we opt for one of the true readings.

- (37) I'm looking for a friend.
 a. Context 1: I just moved to the city and joined a club.
 b. Context 2: I peek into a room and look around.
- (38) *Suzy picks two cards from separate decks. Each is either 1, 2 or 3.*
 If card B were lower than card A, there is a 50% chance that Suzy would have won.

Candidate similarity order:

$$(2, 2) <_w (1, 2), (3, 2), (2, 1), (2, 3) <_w (1, 1), (1, 3), (3, 1), (3, 3)$$

- (39) If card A were 3, Suzy would have won.
 (40) If card A were odd, there is a 50% chance that Suzy would have won.

5. The required de re propositions might not exist.

Idea: *A* mistakenly believes *p* iff (i) *A* believes *p* and (ii) *p* is false.

- (41) *John believes that Alice is in the rain and Bob is dry. In fact both are in the rain (Figure 8).*
 John mistakenly believes that if Alice were near Bob, Alice would be dry.

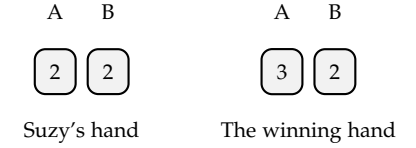


Figure 7:



Figure 8:

For each of John's belief worlds w' at w , if p were true, Alice would be dry.

- De re, anchored to the actual world:
 $p = \{w'' : \text{Alice's location in } w'' \text{ is near Bob's location in } w\}$
 False: John doesn't believe that if Alice were where Bob actually is (location 2), Alice would be dry.
- De re, anchored to John's belief worlds:
 $p = \{w'' : \text{Alice's location in } w'' \text{ is near Bob's location in } w'\}$
 False: it isn't a mistake that if Alice were near where John believes Bob to be (location 3), Alice would be dry.
- De dicto:
 $p = \{w'' : \text{Alice's location in } w'' \text{ is near Bob's location in } w''\}$
 True: John believes that if p were true, they would both be dry, but p is false (if Alice were near where Bob actually is, Alice wouldn't be dry).

While (41) is true de dicto, (42) seems to have no true reading.

- (42) John mistakenly believes that if Bob were near Alice, Alice would be dry.

Variation in counterpart strictness

- (2) a. If New York City were in Georgia, then New York City would be in the South.
 b. If Georgia included New York City, then New York City would be in the South.

We can explain the simultaneous truth of Goodman's sentences [...] by the hypothesis that both are *de re* both with respect to 'New York City' and with respect to 'Georgia', and that a less stringent counterpart relation is summoned up by the subject terms 'New York City' in [(2a)] and 'Georgia' in [(2b)] than by the object terms 'Georgia' in [(2a)] and 'New York City' in [(2b)]. Then in [(2a)] we are concerned with the closest worlds to ours where a not-too-close counterpart of our New York is in a close counterpart of our Georgia, and hence is in (a counterpart of?) the South; whereas in [(2b)] we are concerned with the closest worlds to ours where a not-too-close counterpart of our Georgia includes a close counterpart of our New York City, and hence is not entirely included in the South.

(Lewis 1973a:43)

We certainly *can* explain it this way, but *should* we?

Drawbacks:

1. Does not account for (13) and (38).
2. Faces objection 4 to the *de re* reply (lack of charitable interpretations, even given the flexibility of the counterpart relation).
3. Not expected given independently plausible constraints on the counterpart relation:

something has for counterparts at a given world those things existing there that resemble it closely enough in important respects of intrinsic quality and extrinsic relations, and that resemble it no less closely than do other things existing there (Lewis 1973a:39)

Heller (2005), Stalnaker (1986), and Fara (2009) criticise the analysis of the counterpart relation in terms of similarity.

4. Unsatisfying: "I cannot say in general how grammar and context control which counterpart relation is used where."
5. A general worry: it doesn't extend to other counterexamples to Substitution, blocking a unified approach to Substitution's challenges.

(Lewis 1973b:436)

An aboutness theory of counterfactuals

1. Pick a time at which to imagine the antecedent true.
2. Allow the part of the world the antecedent is about at that time to vary.
3. Play forward the laws.
4. Restrict to worlds where the antecedent is true.
5. A counterfactual ‘if *A*, would *C*’ is true just in case *C* is true at all/the selected resulting world(s).

See McHugh (2022, 2023: chapter 2) for a formalisation.

The subject default. By default, sentences are about their subjects and not about their objects.

Recall (2). Analysis: “New York City is in Georgia” is about New York City (specifically, its location) and not about Georgia, while “Georgia includes New York City” is about Georgia’s location (and not about New York City).

For a more challenging case, recall (13). “*A* is to the right of *A* and *B*’s midpoint” is about where *A* is, and not about where *B* is. We interpret “*A* and *B*’s midpoint” de dicto, so it ends up where it would be were *A* to the right of it.

Advantages over competing approaches.

- (i) *Truthmakers*. Looks at the sentence’s grammatical form, without requiring particular states (such as the relational state of *A* being on top of *B*, which is not also the state of *B* being beneath *A*).
- (ii) *The de re reply*. The aboutness approach invalidates Substitution even when all terms are read de dicto, thus overcoming the challenges facing the de re reply.
- (iii) *Variation in counterpart strictness*. Unlike the counterpart reply, it does not require a modified counterpart relation, and extends to the other counterexamples to Substitution we considered.

The default status of the subject default

Some nuance in what counts as a subject. For the purposes of counterfactual interpretation, the subject in the sentences below is Alice, rather than Bob.

- (43)
 - a. If Alice were next to Bob
 - b. If Bob had Alice next to him
 - c. If Bob were such that Alice were next to him
- (44)
 - a. If Alice hugged Bob, he would feel butterflies in his stomach.

- b. If Bob were hugged by Alice, he would feel butterflies in his stomach.

Sometimes, a sentence can be about its object, as in (45).

- (45) If 16 were the legal drinking age, Alice could order wine.

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